Ch 2.2: Assessing Model Accuracy

Lecture 3 - CMSE 381

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::
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Announcements

Last time:

- Ch 2.1, Vocab day!

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- First homework due Sunday, 9/8. Covers:
  - Mon 8/26 lecture
  - Weds 8/28 Lecture
  - Today 9/4 Lecture
- Office hours
  - Mondays, Dr Munch (Wells TBD) 11am - noon
  - Tuesdays, Christy (EGR TBD) 8:30pm - 10am
  - Thursdays, Christy (EGR TBD) 1 - 2:30pm
  - Fridays, Dr Munch (EGR 1511) 11am - noon

Covered in this lecture

- Mean Squared Error (regression)
- Train vs Test
- Bias Variance Trade off

Quick review of notation

Input variables: $X = (x_1, \ldots, x_p)$

↳ # of input variables

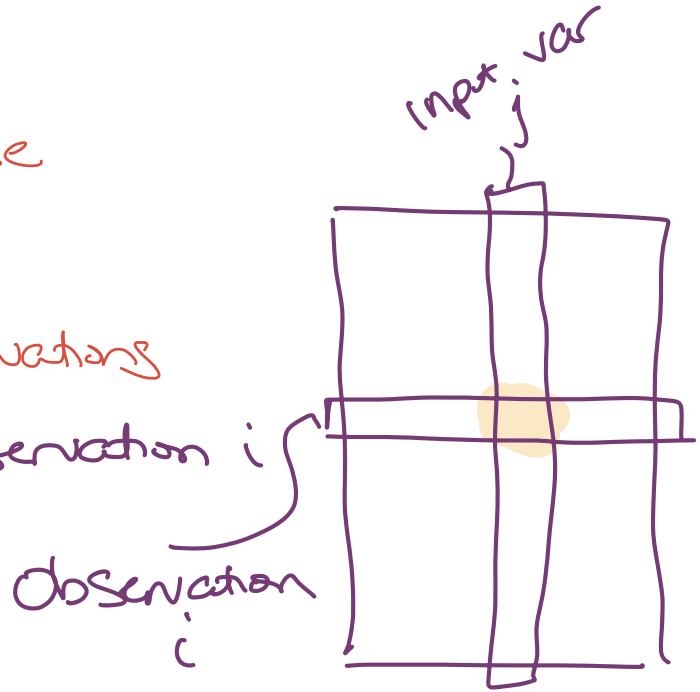
Ground truth $Y = f(x) + \varepsilon$

↳ noise

Approximate $\hat{Y} = \hat{f}(x)$

# of data pts n ← # of observations

x_{ij} is the jth predictor for observation i

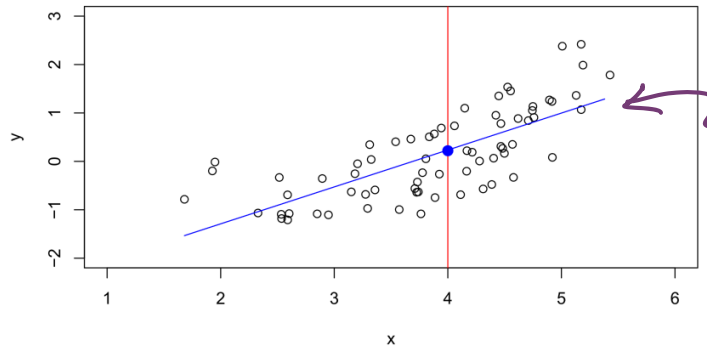


Section 1

Mean Squared Error

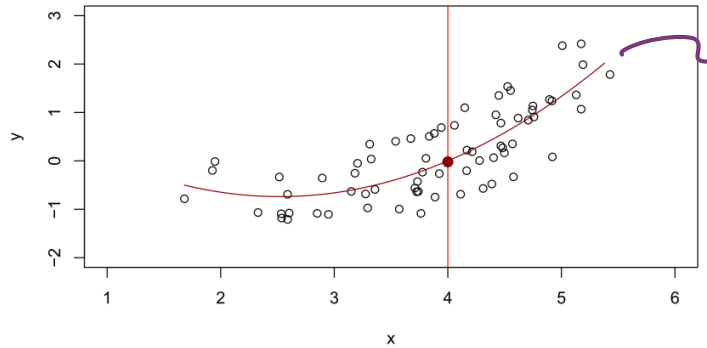
Which is better?

A linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ gives a reasonable fit here



$\hat{f}_L$

A quadratic model $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$ fits slightly better.



$\hat{f}_Q$

No free lunch

- No best model for all datasets
- <u>So</u>

  Need to measure the quality of any model on a given data set.

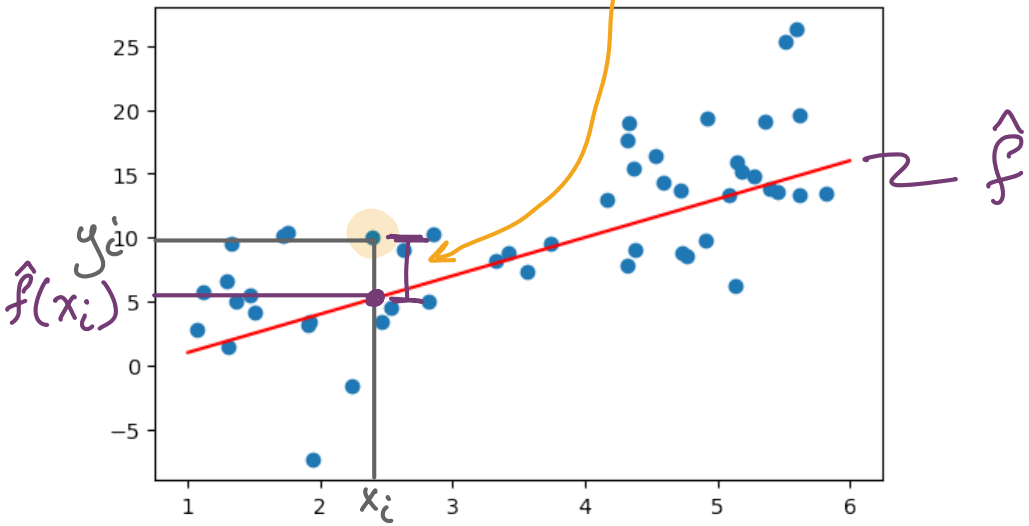
Mean Squared Error

Error in the regression setting

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

observations

observation (x_i, y_i)



Group Work

Given the following data, you decide to use the model

$$\hat{f}(X_1, X_2) = 1 - 3X_1 + 2X_2.$$

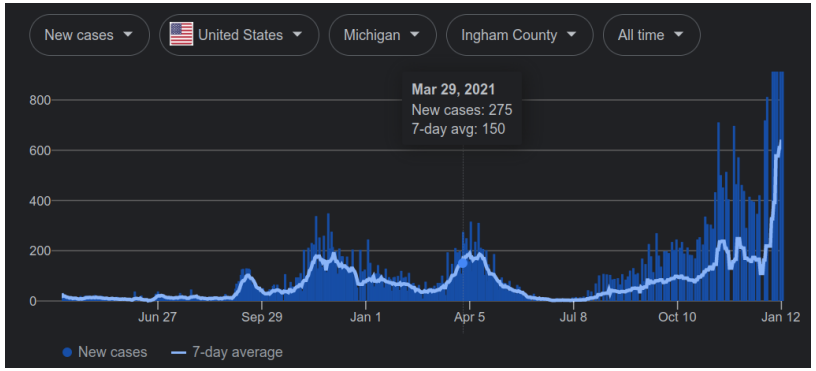
What is the MSE?

X_1	X_2	Y
0	7	14
1	-3	-6
5	2	-10
-1	1	7

$\hat{f}(x_1, x_2)$	$\lvert y_i - \hat{f}(x_1, x_2)\rvert$	$(y_i - \hat{f}(x_1, x_2))^2$
15	1	1
-8	2	4
-10	0	0
6	1	1

$$MSE = \frac{1}{4}(1+4+0+1) = \frac{6}{4} = \frac{3}{2} = 1.5$$

Training MSE



We don't care how well you predict data already saw....

We want to do a good job on data we don't have yet!

Train vs test

Training set:
The observations
$\{(x_1, y_1), \cdots, (x_n, y_n)\}$ used to get the estimate $\hat{f}$

Test set:
The observations
$\{(x'_1, y'_1), \cdots, (x'_{n'}, y'_{n'})\}$ used to compute the average squared prediction error

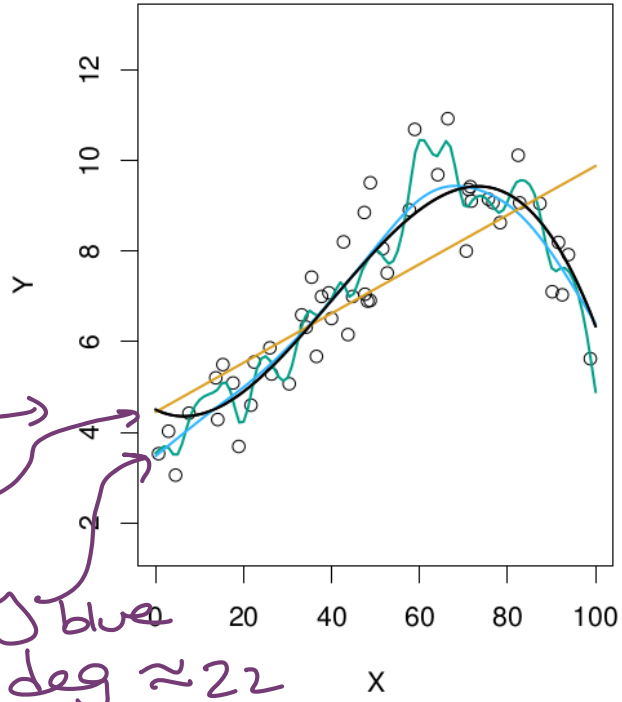
$$\frac{1}{n'} \sum_i (y'_i - \hat{f}(x'_i))^2$$

data not used for training the model

"Data Snooping"
Used the test set data to train

Why not just get the best model for the training data?

$Y = f(X) + \varepsilon$

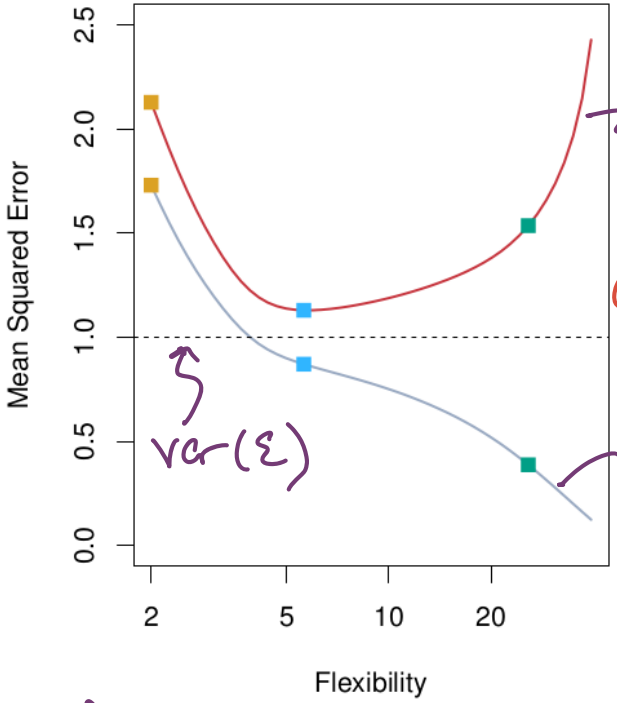


modeled using linear gold

deg 5 poly blue

green deg ≈ 22

Black line: f, used to simulate my data.



var(ε)

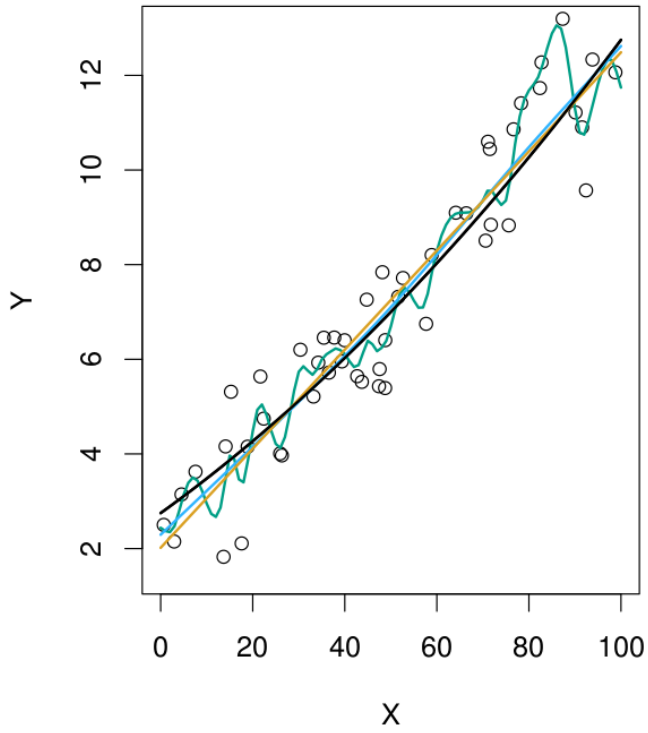
MSE on unseen data

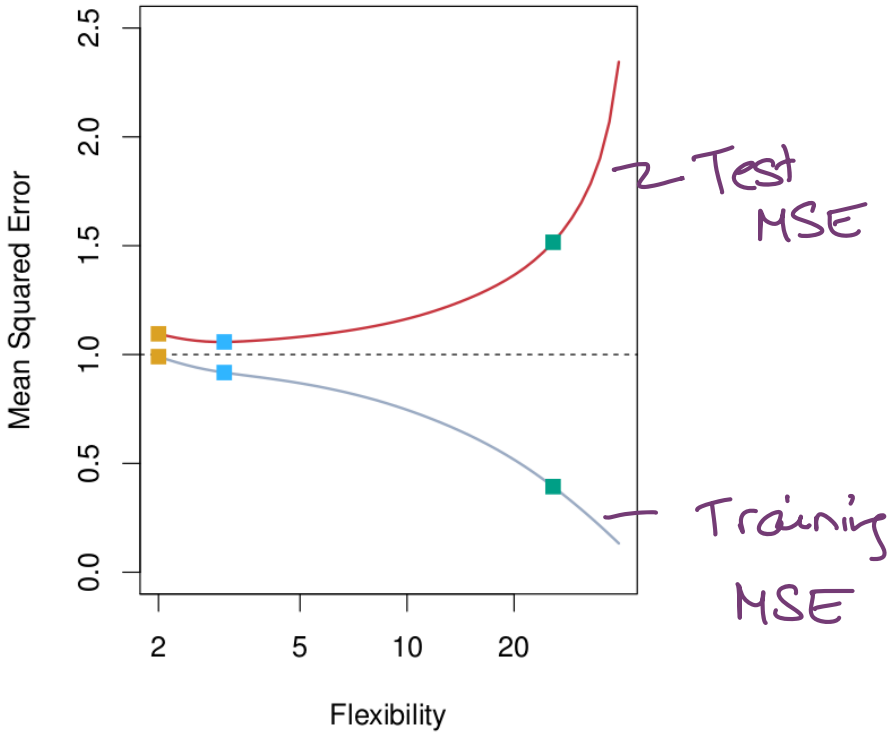
Overfitting!

MSE of each model on the training data

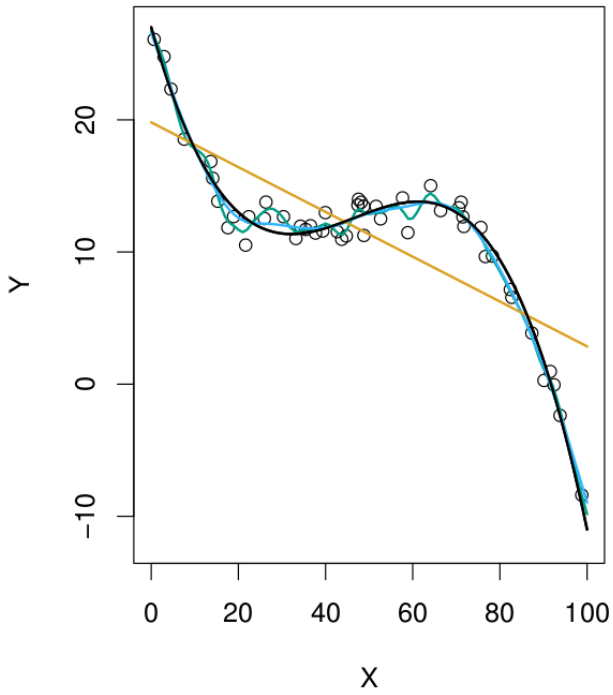
MSE down but only on input

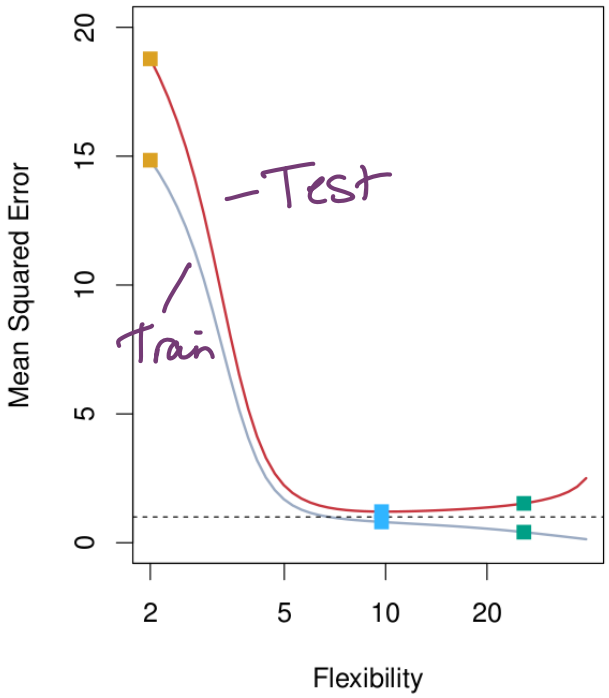
A more linear example





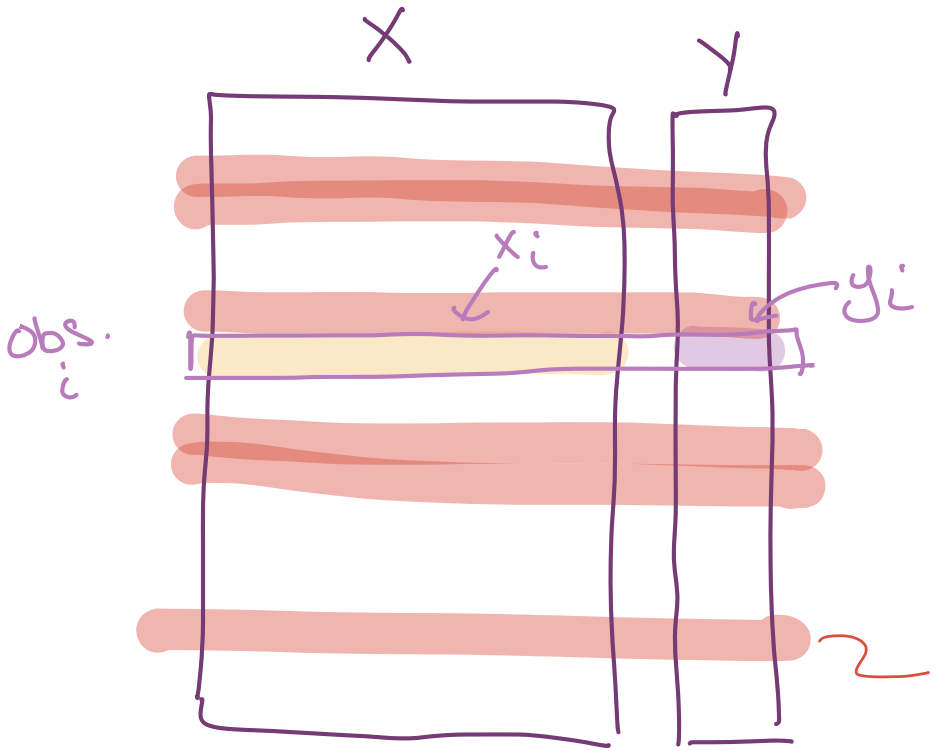
A more non-linear example





A simple solution: Train/test split

More on this in Ch 5



Pick some proportion $q \in [0,1]$

Take q % of the data (randomly select)

Hold out for testing later

Train on the rest

Section 2

Bias-Variance Trade-Off

Bias-variance

$$\underbrace{E(y_0 - \hat{f}(x_0))^2}_{\text{Expected val of testing MSE}} = \underbrace{\mathrm{Var}(\hat{f}(x_0))} + \left[\underbrace{\mathrm{Bias}(\hat{f}(x_0))}\right]^2 + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{Irred. error}}$$

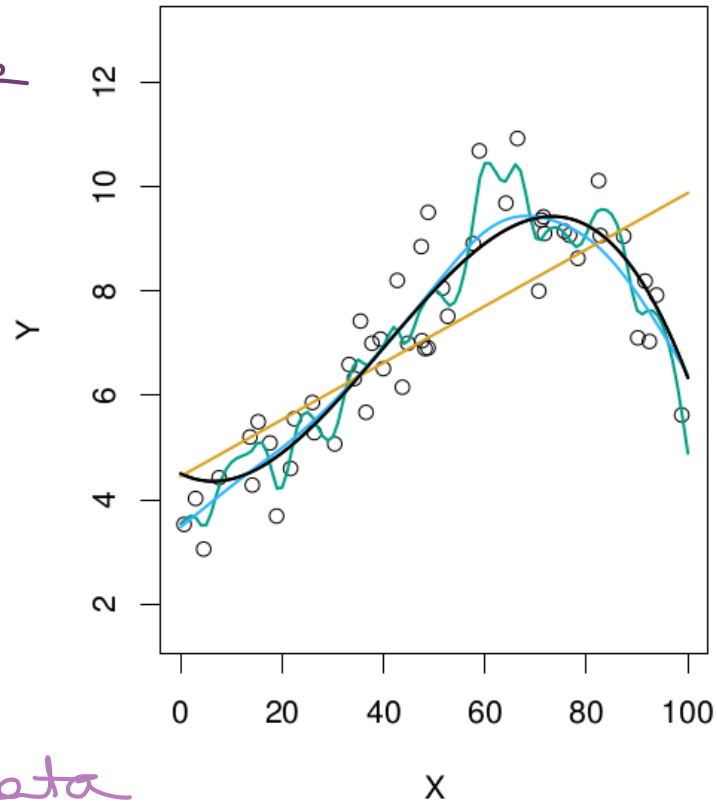
avg test MSE if repeatedly estimated f with lots of training sets and tested each @ x_0

- Proof outside scope of class
- If $E(y_0 - \hat{f}(x_0))^2$ to be low, want everything on the right to be small

Variance

High variance:
Small changes in training data result in large changes in $\hat{f}$

Variance: the amount by which $\hat{f}$ would change if we estimated it using a different training data set.

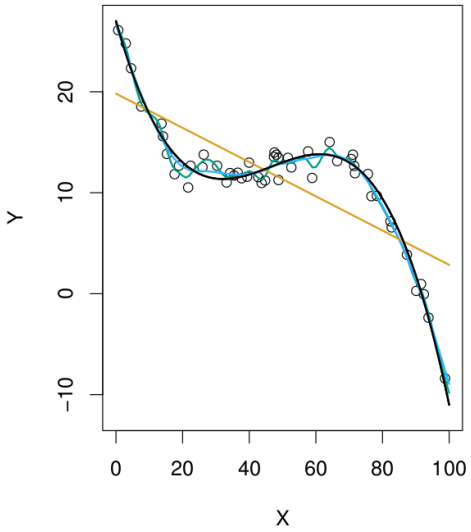


In this example,
green $\hat{f}$, more flexible
but also small changes in data result in very different wiggles

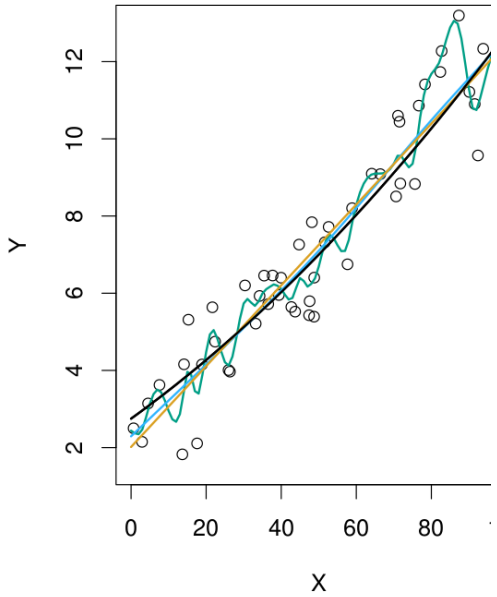
Bias

Bias: the error that is introduced by approximating a (complicated) real-life problem by a much simpler model.

Ex) linear regression often too simple for any real life problem

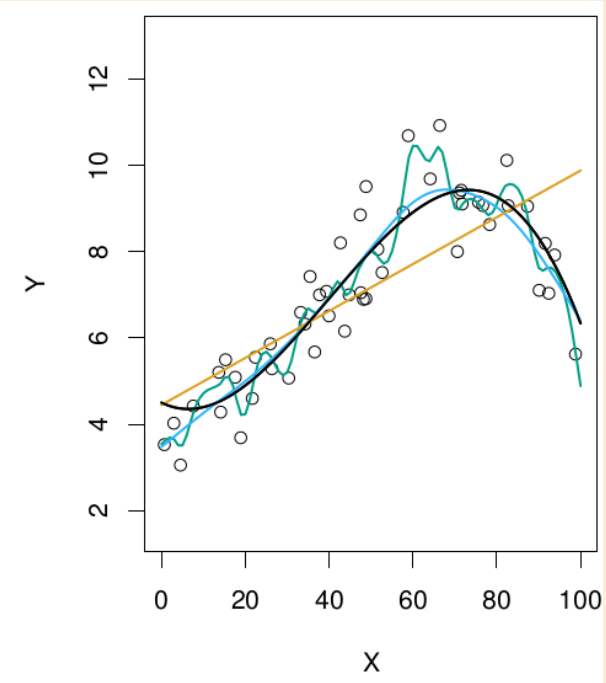


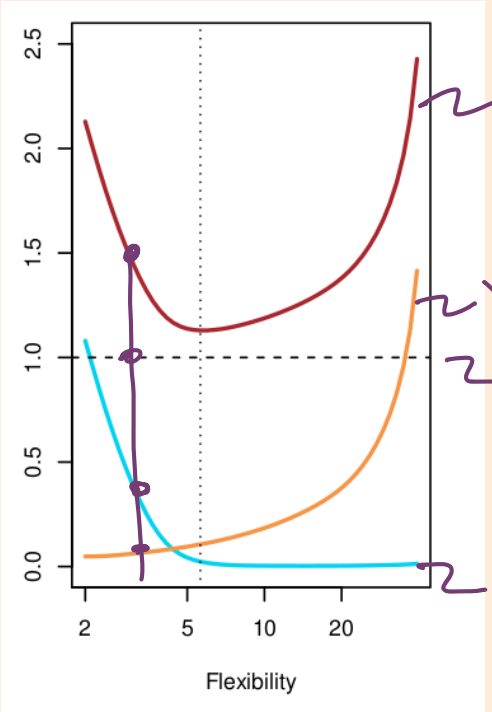
True f is non-linear
No good lin. model
lin reg = high bias



True f is close to linear
lin reg = low bias

Group work



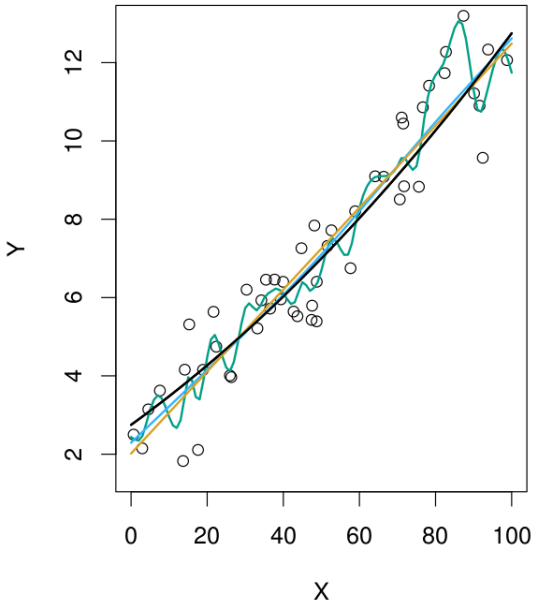


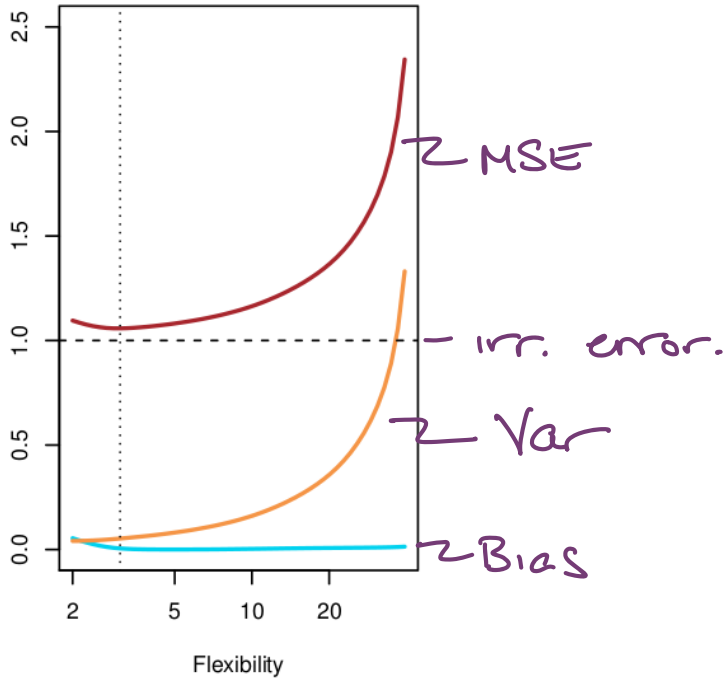
Label the line corresponding to each of the following:

- MSE – Test
- Bias
- Variance of $\hat{f}(x_0)$
- Variance of ε

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

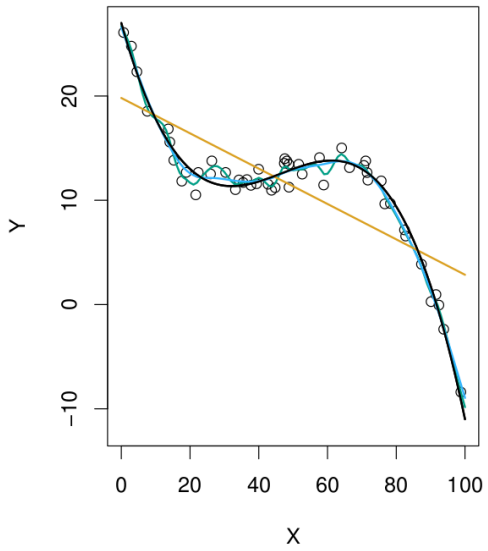
Another example

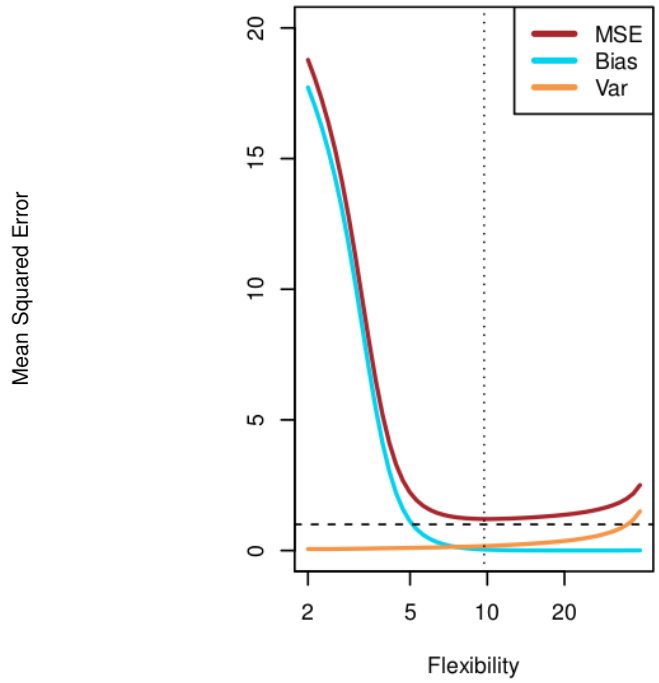




$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

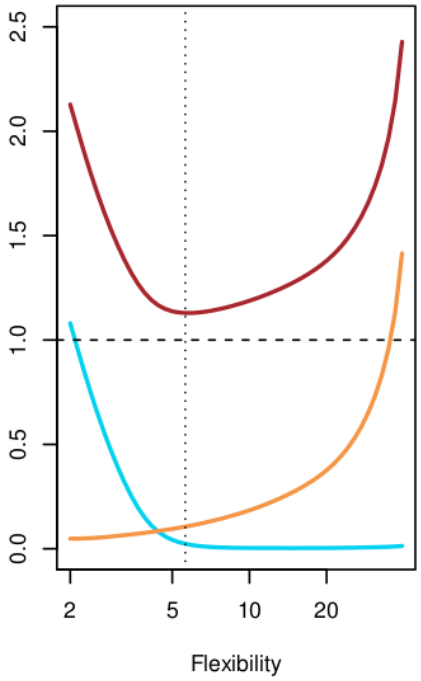
Yet another example





$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Bias-variance trade off



- More flexible model:
  - Variance ↑
  - bias ↓
- Rel. rate of change of the two determines whether MSE up or down
- In general, looking for the min of MSE

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Group work: coding

See jupyter notebook

Next time

- Friday:
  - 3.1 Linear Regression
- Sunday
  - Homework due midnight on D2L

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
	Fri	8/30	Class Cancelled (Dr Munch out of town)		
	Mon	9/2	No class - Labor day		
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1/3.2	